

Evaluating the effects of NMN on Amnesiac *Drosophila melanogaster*

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Introduction

Amnesia is a condition that affects millions every year (Mayo, 2022). This disease is very crippling for both the individual and the families of these individuals. Amnesia is characterized by the sudden loss of memory in an individual and in some cases prevents them from creating new memories (Mayo, 2022). There is currently no reliable treatment for people with amnesia (Mayo, 2022). Research regarding treatments for amnesia needs to be furthered. One chemical that may prove to be an effective remedy is Nicotinamide Mononucleotide (NMN). NMN is used by the body to synthesize Nicotinamide adenine dinucleotide (NAD⁺). NAD⁺ then reduces oxidative stress within the body by activating the Nrf2/Keap1/NQO1 pathways which increases the production of Nrf2 protein, decreases the production of Keap1 protein, and increases the antioxidant factor of NQO1 (Ma et al., 2024). These factors are linked to reduced oxidative stress. This can also happen in the brain which would affect neural pathways and hinder memory development and memory retention (Singh et al., 2022). NMN could be a probable treatment for reducing oxidative stress and, by extension, amnesia.

Purpose

There is a lack of clear understanding of the benefits of Nicotinamide Mononucleotide (NMN) on human memory. This should thus be studied as there is a precedence for NMN to aid in the treatment of neurodegenerative diseases such as Alzheimer's and Huntington's. This may translate to memory loss and aid with poor memory. For this reason, it should be studied to see if there is a significant increase in memory capabilities due to NMN. Thus, this study aims to further the understanding of NMN on memory.

Method

Toxicology Assay: *Drosophila* at 3-4 days old were selected. All *Drosophila* were fed their respective concentrations of NMN. They were kept in the vials of food for multiple days to ensure feeding occurs. Five flies were placed in each vial, and the amount that were alive on a date was recorded for 6 class periods to ensure adequate data collection.

APS Assay: *Drosophila* at 3-4 days old were selected. All *Drosophila* were fed their respective concentrations of NMN. They were kept in the vials of food for multiple days to ensure feeding occurs. *Drosophila* were placed into a T-maze after being anesthetized and sexed using cold sorting. One end had light and quinine solution, the other side was in darkness to induce negative phototaxis behavior. This will teach flies to go away from the light. Flies were retested after 3 hours and the results were recorded and analyzed.

Toxicology Assay

Date	Male Concentrations (mM)					Female Concentrations (mM)				
	0	1	5	10	0	1	5	10	0	1
20-Feb	5	5	5	5	5	5	5	5	5	5
24-Feb	5	5	5	5	5	5	5	5	5	5
2-Mar	5	5	5	5	5	5	5	5	5	5
4-Mar	5	5	5	5	5	5	5	5	5	5
6-Mar	5	5	5	5	5	5	5	5	5	5
10-Mar	5	5	5	5	5	5	5	5	5	5

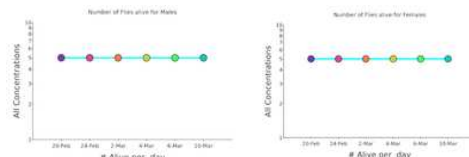


Figure 2. Data table and scatterplot of the survival counts for each concentration of NMN at each gender

The data shows that there is a trend of safety among the concentrations of NMN. The populations have remained stable from the data recorded and has not seen death among the *Drosophila* for reasons regarding the drugged feed. As a result the APS assays can continue with these concentrations safely.



Figure 1. T-Maze for the APS assay.

APS Assay

Effect	Degrees of Freedom	F statistic (adj. for residual)	P-value	Interpretation of P-value
Concentration of NMN (adj)	3	2.4	0.11	P-value is > 0.05, no significant difference between groups
Error or Residual	16			

Effect	Degrees of Freedom	F statistic (adj. for residual)	P-value	Interpretation of P-value
Concentration of NMN (adj)	3	0.39	0.76	P-value is > 0.05, no significant difference between groups
Error or Residual	16			

Figure 3. Data Table for the pass count for each fly zero and three hours post conditioning, with an ANOVA analysis for both genders from DataClassroom. (Male on the top, Female below)

An ANOVA test was ran on the difference in pass counts between zero and three hours after conditioning the flies. It was found that there was no difference due to a high p-value for both females and males, when compared with a significance level of 0.05.

Conclusion

The conclusion of the Toxicology assay, due to the data showing no deaths, can be that there was no danger observed for the concentrations of NMN administered on the amnesiac flies. However, the APS assay shows that there was no difference in the flies' pass rate after taking NMN. This was seen across both genders and indicates that NMN did not have a significant impact on the memory of the flies from the APS Assay. A likely reason for this conclusion could have been that the NMN's effect on oxidative stress did not affect the memory of the flies. Likely this pathway may have not been linked to how the amnesia in the flies deteriorated the memory. Future research could possibly focus on employing a different method of memory loss other than a genetic inhibition. This could include a chemical designed to cause memory loss in a wild-type fly, or a different assay for memory that does not target phototoxic memory. As a result, further testing with another set of parameters should be utilized to help ameliorate the issue of amnesia utilizing a drug-based treatment.

References

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